



North South University

Department of Electrical & Computer Engineering

Lab Assessment

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Lab No:	03
Title:	Conditional Branching in ARMv7 Assembly
Course Code:	CSE331L.04
Course Name:	Microprocessor Interfacing & Embedded System Lab
Section:	Section 04
Faculty:	Mr. Md Raqibur Rahman [RQN]
Date of Submission:	01st March, 2026
Due Date of Submission:	01st March, 2026
Submitted to:	Md. Wardun Islam Noon

Question:

Task 1:

Check if a number is even or odd, store 1 in a register if even otherwise 0

Task 2:

Use loop to calculate sum of the following series $1+2+3+\dots+10$

Solution:

```
1. .global __start
   __start:

       mov r0,#0
       mov r1,#10
       mov r3,#1
       mov r4,#0

       and r1,#1

       cmp r1,r3

       beq body
       b else

       body:
       mov r0,#1
       b exit

       else:
       mov r0,#0

       exit:

2.

.global __start
__start:

       mov r0,#0
       mov r1,#1
       mov r2,#10

       loop:
       cmp r1,r2
       ble body
       b exit
```

```
body:  
add r0,r0,r1  
add r1,r1,#1  
b loop
```

```
exit:
```